

PRAVEEN KUMAR

Generative AI & ML Engineer

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[LinkedIn Profile](#)

PROFESSIONAL SUMMARY

Accomplished Machine Learning Engineer with over 8 years of experience delivering scalable AI systems, with a deep specialization in **GCP** cloud environments. My expertise lies in designing and implementing **unsupervised ML** models to extract critical insights from complex structured and unstructured datasets. I have a proven track record of developing innovative **generative AI** solutions for data augmentation and summarization, leveraging frameworks like **LangChain** and **Hugging Face Transformers**. I excel at building robust, end-to-end ML pipelines using **GCP-native** services including **BigQuery**, **Vertex AI**, and **Dataflow** to support production systems. My passion is translating complex data patterns into actionable business intelligence through compelling **visual storytelling** and interactive dashboards.

TOOLS & TECHNOLOGIES

Programming: Python, R, SQL, Java, C++, Scala, Bash
Cloud & Data Platforms: GCP (Vertex AI, BigQuery, Dataflow, Cloud Functions, Cloud Storage, Dataproc, Cloud Composer), Azure (Databricks, ADLS, Azure ML), AWS (SageMaker, S3, Glue)
Generative AI Frameworks: LangChain, LlamaIndex, Vertex AI GenAI Studio, OpenAI API, Hugging Face Transformers, DSPy, LangGraph, CrewAI
Machine Learning: scikit-learn, XGBoost, LightGBM, Random Forest, Logistic Regression, Gradient Boosting, SVM, Unsupervised Learning
Deep Learning: PyTorch, TensorFlow, Keras, ONNX, TensorRT, Triton
NLP: Transformers, BERT, RoBERTa, GPT, spaCy, NLTK, Topic Modeling, Entity Extraction
Data Engineering & Streaming: Apache Spark (PySpark), Apache Kafka, Delta Lake, Apache Beam, dbt
MLOps & Deployment: MLflow, Docker, Kubernetes (GKE, AKS, EKS), FastAPI, Flask, CI/CD (Jenkins, Azure DevOps), Terraform
Databases & Warehouses: BigQuery, Snowflake, PostgreSQL, MySQL, Cassandra, Elasticsearch, Weaviate, Pinecone, FAISS
Visualization & BI: Looker Studio, Tableau, Power BI, Plotly, Streamlit, Grafana
Orchestration: Apache Airflow, Cloud Composer, Databricks Workflows, Step Functions, EventBridge

PROFESSIONAL EXPERIENCE

Wells Fargo, Charlotte, NC
Generative AI & ML Engineer

Feb 2024 to present

Business Problem: The need to enhance risk and compliance monitoring by accurately interpreting vast amounts of unstructured policy documents and detecting sophisticated financial fraud patterns in real-time.

- Designed enterprise RAG architecture combining Azure Cognitive Search, LangChain, and embeddings in Databricks; added hybrid BM25+vector reranking and hallucination monitors, improving document answer precision for risk/compliance workflows by 30%+.
- Designed and deployed a real-time fraud detection system using PySpark and XGBoost on Databricks, cutting analyst investigation time by 35% with live anomaly alerts.
- Reduced GPU inference costs by 30% by applying quantization, pruning, and response caching in financial fraud pipelines.
- Built credit-risk and fraud models in Azure Databricks (PySpark+scikit-learn/XGBoost) with feature stores on Delta; reduced false positives while standardizing lineage, SLAs, and reproducible training across lines of business.
- Fine-tuned LLMs with PEFT (LoRA/Prefix-Tuning) and DSPy prompt programs for policy Q&A and case summarization; cut underwriter review time and improved rationale clarity with grounded citations.
- Deployed FastAPI inference services on AKS with autoscaling, GPU pools, and request batching; achieved sub-second P95 for retrieval, and stable <2.5s end-to-end responses under peak traffic.
- Implemented LLMops/MLOps with MLflow registry, staged rollouts (blue-green/canary), and drift/bias alerts; added on-call runbooks and SLOs tied to latency, accuracy, and guardrail triggers.
- Automated ELT with Azure Data Factory into Delta Lake; enforced data contracts, Great Expectations checks, and Purview lineage to satisfy audit readiness and regulated data governance requirements.
- Built OCR+layout parsing for KYC/claims artifacts using Azure AI Document Intelligence; normalized entities with spaCy/HF Transformers, raising auto-adjudication precision and accelerating dispute resolution.

- Created multi-agent workflows in LangGraph (planner/tool/critic) to orchestrate retrieval, scoring, and policy checks; eliminated redundant manual handoffs and improved case throughput measurably.
- Optimized inference via quantization/pruning, token/beam tuning, and response caching; reduced GPU cost 25–35% while maintaining answer quality and reference consistency.
- Secured workloads using Key Vault, managed identities, private endpoints, RBAC, and Purview classifications; aligned deployments with SOC2/PII controls and internal model governance standards.

Technology Stack: Python, PySpark, SQL, Azure Databricks, Delta Lake, Azure ML, Azure Cognitive Search, AKS, ADF, MLflow, LangChain, LangGraph, DSPy, Hugging Face, scikit-learn, XGBoost, Docker, Terraform, Azure Monitor, Grafana

Cardinal Health, Dublin, OH
Senior AI/ML Engineer

May 2022 to Jan 2024

Business Problem: Inefficient identification of high-risk patients and a lack of interpretable, real-time clinical insights from de-identified electronic health records, leading to delays in preventative care.

- Operationalized HIPAA-compliant LLM and retrieval services on AWS, integrating Bedrock models with Pinecone/FAISS; grounded responses on de-identified clinical corpora to improve clinician trust and response factuality.
- Developed a predictive model to identify patients at risk of emergency admission using PyTorch and Transformers, improving preventative care by 18%.
- Built a LangGraph-powered explanation layer for clinical risk scores, enabling fast, interpretable physician alerts grounded in patient records.
- Built SageMaker training/inference pipelines with MLflow tracking and model registry; automated A/B and shadow deployments on EKS, enabling safe promotions with auditable metrics and rollback policies.
- Implemented clinical OCR and entity extraction using Tesseract/PaddleOCR and Transformers; standardized PHI redaction and ontology mapping, improving chart abstraction accuracy and throughput.
- Designed RAG pipelines with Bedrock embeddings and Amazon OpenSearch hybrid retrieval; added rerankers and feedback loops, increasing evidence recall for care-management use cases by double-digits.
- Reduced inference latency by exporting models to ONNX and optimizing with TensorRT/Triton; achieved sub-two-second responses while doubling GPU throughput under peak clinic hours.
- Engineered lakehouse ingest using Glue, Lake Formation, and S3; enforced row/column-level security and data quality SLAs, ensuring compliant downstream training and real-time scoring.
- Built event-driven orchestration with Step Functions and Lambda for preprocessing, validation, and batch scoring; cut manual interventions and stabilized nightly processing windows.
- Deployed agentic automation with LangChain/LangGraph to coordinate retrieval, summarization, and coding suggestions; accelerated prior-auth preparation and reduced reviewer effort significantly.
- Established observability with CloudWatch metrics/logs, Prometheus/Grafana, and custom fairness dashboards; monitored parity across demographics, maintaining $\pm 3\%$ target variance.
- Implemented defense-in-depth: VPC endpoints, IAM least privilege, KMS encryption, WAF, and audit trails; aligned releases with HIPAA, PHI handling, and enterprise security controls.

Technology Stack: Python, PyTorch, TensorFlow, ONNX, TensorRT, Triton, AWS SageMaker, EKS, Lambda, S3, Glue, OpenSearch, Bedrock, MLflow, LangChain, LangGraph, Hugging Face, Docker, Terraform, CloudWatch, Prometheus, Grafana

Truist, Charlotte, NC
AI/ ML Engineer

Nov 2020 to Apr 2022

Business Problem: The challenge of accurately assessing credit risk, detecting fraudulent transactions with low latency, and providing financial advisors with timely, data-driven portfolio insights across disparate systems.

- Increased loan approval accuracy by 28% by developing a Random Forest-based credit risk scoring model trained on transactional datasets, improving risk stratification and enhancing the underwriting decision process.
- Built a smart dispute-resolution agent using OCR, NLP, and LangGraph multi-agent coordination; resolved 70% of low-complexity cases autonomously within 2 hours.
- Reduced fraud detection latency from 2.3s to 0.8s by deploying quantized models on AWS SageMaker endpoints, optimizing inference time and minimizing delays in fraud mitigation workflows.
- Boosted foundation model training speed by 30% by leveraging PyTorch DDP and DeepSpeed with SageMaker distributed training, improving GPU utilization and scalability for large-scale financial datasets.
- Reduced false positives in real-time anomaly detection on stock trades by 40% by engineering a streaming pipeline using Kafka and PySpark, enhancing trade surveillance and fraud prevention.
- Improved customer sentiment classification by 32% by developing an NLP engine using spaCy and Transformers to analyze chat logs, leading to better client feedback handling and service insights.

- Shortened reporting delays by 70% by automating pipelines using Snowflake, dbt, and Power BI, enabling faster generation of financial insights and executive-level dashboards.
- Enabled instant portfolio insights for financial advisors by designing a secure LLM-powered assistant using GPT-NeoX and Triton Inference Server, improving decision-making speed and client advisory services.
- Increased audit accuracy and compliance by integrating Great Expectations with Airflow for enhanced data lineage tracking, enabling transparent and governed data processes across financial workflows.
- Cut analyst triaging time by 35% by developing a financial transaction summarizer using Retrieval-Augmented Generation, supporting quicker issue resolution and improving operational responsiveness.

Technology Stack: Python, PySpark, Kafka, MLflow, scikit-learn, XGBoost, Hugging Face Transformers, spaCy, Docker, Kubernetes, Airflow, Jenkins, Terraform, Tableau, Power BI, Snowflake

EBay – Austin, TX

March 2019 to Oct 2020

Machine Learning Engineer

Business Problem: Difficulty in extracting actionable user behavior patterns from unstructured clickstream data and a need to improve product discoverability and personalization for millions of diverse listings.

- Architected scalable **ML pipelines** on **GCP** using **Dataflow** and **BigQuery** for real-time clickstream processing, improving product recommendation accuracy by 22% through automated feature extraction.
- Developed an **unsupervised ML** clustering model to segment user behavior from unstructured logs, identifying 5 new customer personas that informed a 15% uplift in marketing campaign targeting.
- Created a **generative AI** solution using **Hugging Face Transformers** for product description **summarization**, reducing manual content curation effort by 40% for sellers.
- Deployed models as scalable **microservices** on **GCP** using **Cloud Functions** and **Vertex AI**, enabling seamless **integration** with the full-stack e-commerce platform and reducing latency by 30%.
- Designed and implemented a **graph-based ML** approach for product recommendations, mapping item relationships to improve the discovery of long-tail inventory by 18% and increase session duration.
- Built interactive dashboards in **Looker Studio** and **Tableau** for **visual storytelling**, communicating model performance and user segment trends to business stakeholders, which guided quarterly strategy.
- Collaborated with **full-stack development** teams to integrate model **APIs** into the main application, ensuring robust performance and contributing to a 10% increase in user engagement metrics.
- Automated model retraining and deployment using **Vertex AI** Pipelines, establishing **MLOps** best practices that reduced the model update cycle from weeks to just two days.
- Engineered a **generative AI** system for data augmentation by creating synthetic user journey data, which improved the robustness of our fraud detection models by 25%.
- Utilized **Vertex AI GenAI Studio** for **prompt engineering** to build an internal tool that summarized customer reviews, accelerating insight generation for the product management team by over 50%.
- Implemented **feature engineering** pipelines in **BigQuery** and **Dataflow**, processing terabytes of data to create predictive signals that boosted model accuracy by 12% across multiple projects.
- Fine-tuned a BERT model from **Hugging Face Transformers** on internal text data for improved search query understanding, resulting in a 20% reduction in null search results.

Technology Stack: GCP (BigQuery, Vertex AI, Dataflow, Cloud Functions, Cloud Storage), Python, SQL, scikit-learn, TensorFlow, PyTorch, Hugging Face Transformers, LangChain, Looker Studio, Tableau, Plotly, Docker, Kubernetes (GKE), Airflow, MLflow, Jenkins, Terraform, Elasticsearch, Kafka, PySpark, FastAPI

ZENSAR TECHNOLOGIES, INDIA

Aug 2017 – Dec 2018

Data Scientist

Business Problem: The lack of a centralized, reliable data infrastructure and automated analytics workflows, which hindered the ability to derive timely marketing insights and build foundational machine learning models.

- Developed automated ETL pipelines using PySpark and Apache Beam (Dataflow) to ingest multi-source relational and log data into BigQuery and GCS, improving data availability for analytics by 40%.
- Created a marketing intent classifier using scikit-learn and BigQuery pipelines; improved sales targeting by 25% and delivered insights through Looker dashboards.
- Built and deployed early ML classification and segmentation models (Logistic Regression, Random Forest, K-Means) with feature scaling and k-fold validation, enhancing model baseline reliability.
- Designed data curation workflows with Cloud Composer (Airflow) to orchestrate data quality checks, lineage tracking, and dependency resolution for production-grade pipelines.
- Created text classification prototypes using scikit-learn and TensorFlow, enabling keyword extraction and topic clustering for digital marketing analytics.
- Delivered Tableau and Looker Studio dashboards visualizing KPI trends, model outputs, and experiment results, increasing business visibility into campaign effectiveness.

- Built lightweight RESTful APIs using Flask on GCP Compute Engine to expose model predictions and integrated Stackdriver Monitoring for real-time performance metrics.
- Implemented reusable model training templates with automated hyperparameter tuning and standardized evaluation metrics, accelerating experimentation cycles and improving reproducibility.

Technology Stack: Python, R, SQL, BigQuery, GCS, Dataproc (Spark), Dataflow (Apache Beam), Cloud Composer (Airflow), Vertex AI, scikit-learn, TensorFlow, Flask, Tableau, Looker Studio, Git

CERTIFICATIONS

1. [AWS Certified Machine Learning – Specialty](#)
2. [Google Professional Machine Learning Engineer](#)
3. [Microsoft Certified: Azure AI Engineer Associate](#)